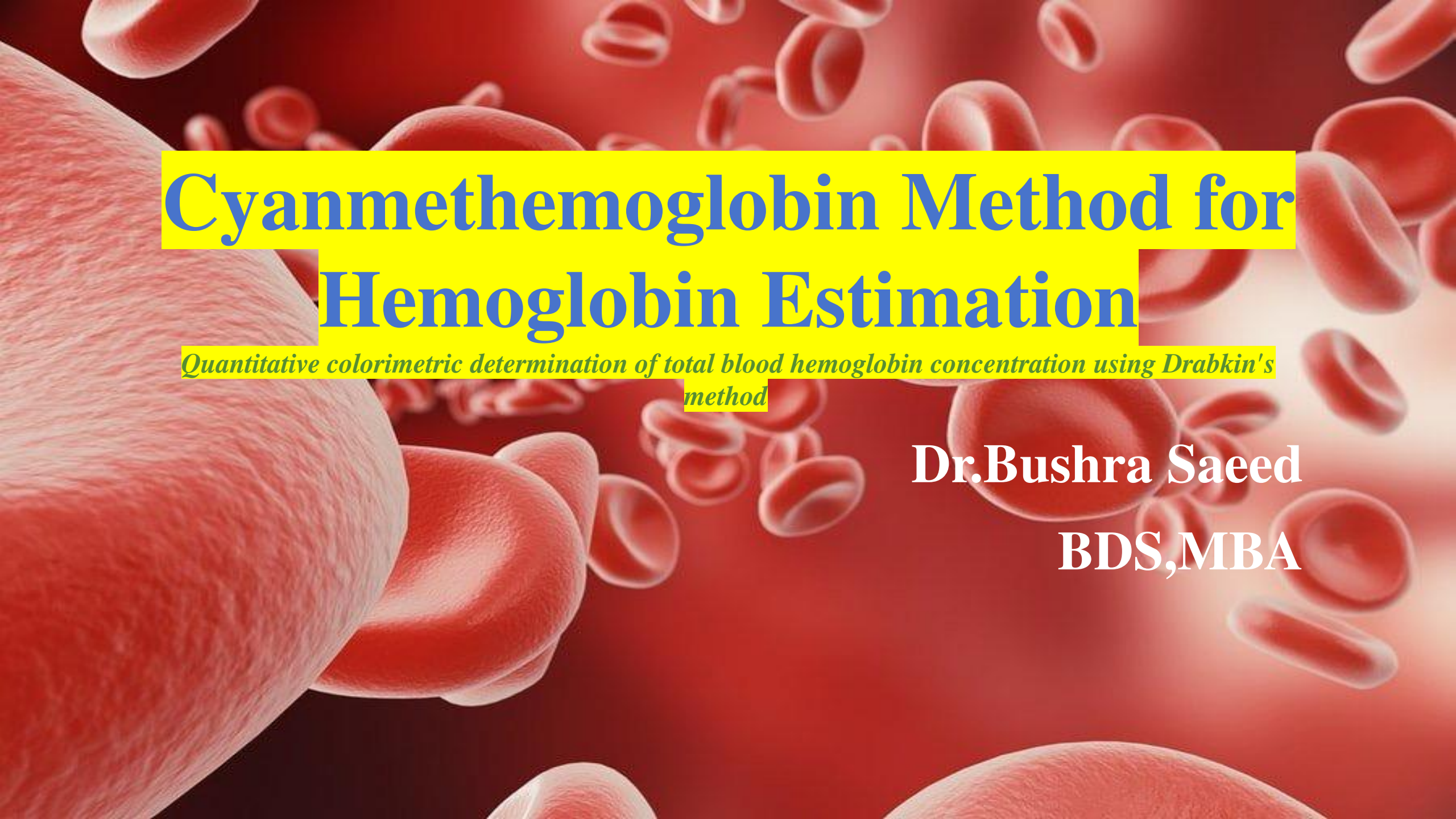


Cyanmethemoglobin Method for Hemoglobin Estimation

Quantitative colorimetric determination of total blood hemoglobin concentration using Drabkin's method

Dr.Bushra Saeed

BDS,MBA

Hemoglobin Role & Reference Values

Oxygen Transport Pigment

Hemoglobin is the red , oxygen-carrying pigment contained in red blood cells that is responsible for delivery of oxygen to the tissues and then transport carbon dioxide out of the tissues back to the lungs.

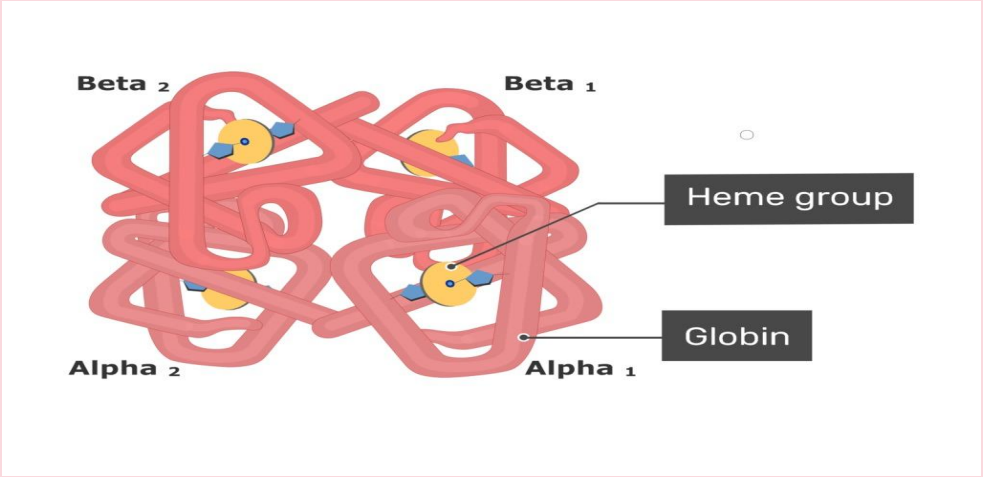
Measured through a standard blood test like a Complete Blood Count (CBC).

Function and Structure

• Oxygen Transport: Each molecule contains four heme groups with iron atoms that reversibly bind up to four oxygen molecules.

• Carbon Dioxide and Buffering: It helps carry a portion of carbon dioxide back to the lungs and balances pH levels in the blood.

• Structure: Composed of four protein subunits (usually two alpha and two beta chains in adults)



Male Reference Range ♂

13 – 18 g/dL

Female Reference Range ♀

12 – 16 g/dL

Laboratory Methods for Hemoglobin Estimation

Various methods for estimation of hemoglobin in the laboratory

1- Methods based on development of color, which include:

a-Sahli method (Acid hematin visual match)

b-Drabkin (Cyanmethemoglobin) method

2- Automated Hematology Analyzer

3- Measurement of oxygen combining capacity by blood gases analyzer (BGAs)

Required Reagents & Equipment

Drabkin's Reagent

Potassium Ferricyanide: Oxidizing agent

Potassium Cyanide: Cyanide reactant

Blood Specimen

Venous Blood: Collected in EDTA tube

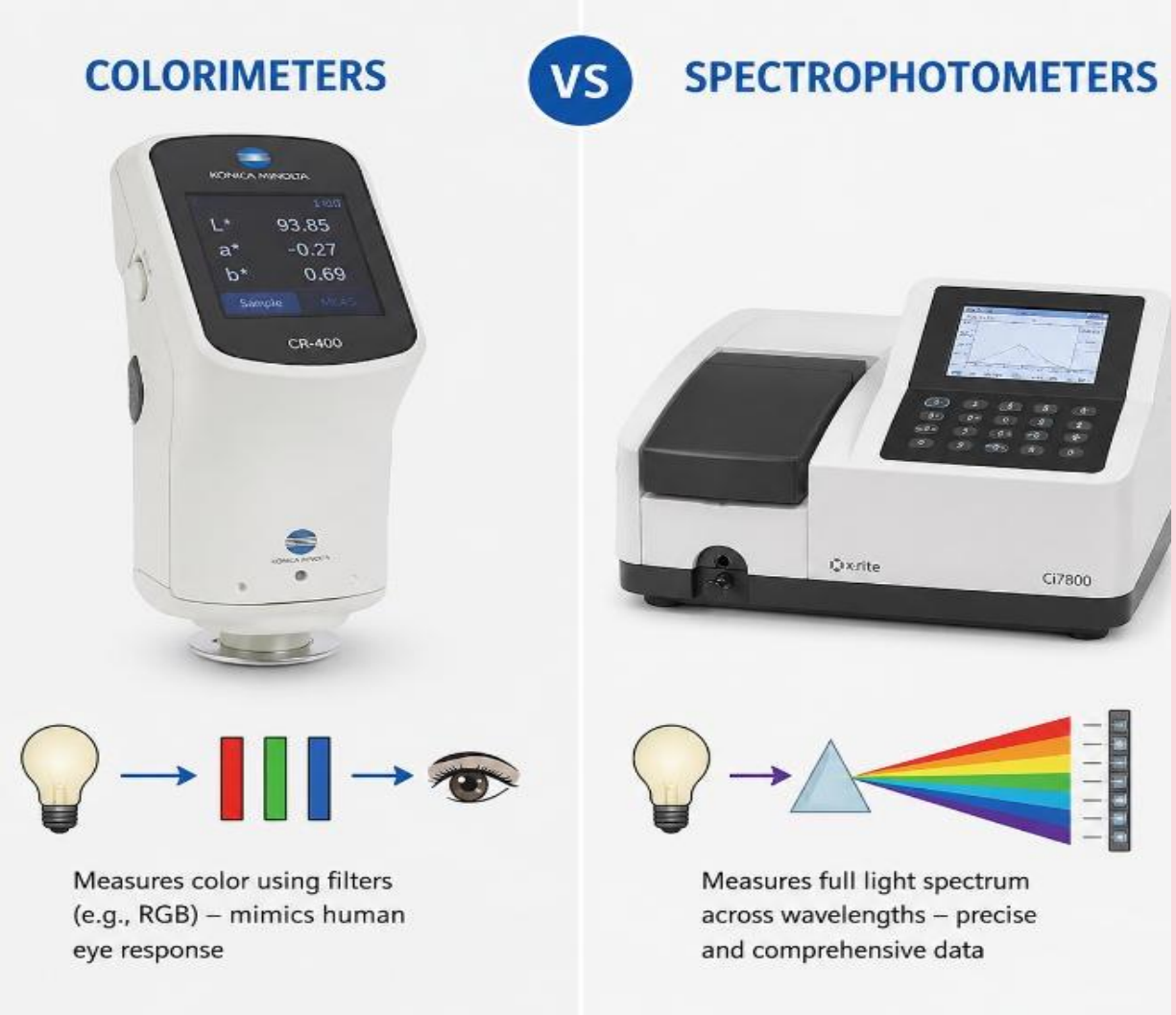
Capillary Blood: Direct puncture sample

Instrumentation

Spectrophotometer / Colorimeter: 540 nm setting

Cuvettes & Pipettes: Precision volumetric tools

Continued...



- **- Colorimeter:** A simple device that measures light absorption using colored filters in the visible range.
- **- Spectrophotometer**: An advanced device that measures light absorption at specific wavelengths using a prism or diffraction grating. It can analyze UV and visible light, and some instruments also measure IR.
- **Key difference:** A colorimeter uses filters and is simpler, while a spectrophotometer provides more precise wavelength selection and broader measurement capabilities

Drabkin (Cyanmethemoglobin) Method

Flowchart: Principle and Procedure

Objective:

To determine the hemoglobin concentration in whole blood using a colorimeter or spectrophotometer at 540 nm.

1 Whole blood

Contains red blood cells (RBCs) and hemoglobin.



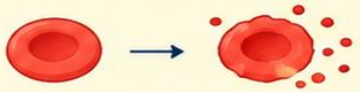
2 Add Drabkin's reagent

Blood is diluted 1:250 with Drabkin's solution (contains potassium ferricyanide and potassium cyanide).



3 Hemolysis of RBCs

Drabkin's solution breaks open RBCs, releasing hemoglobin into the solution.



4 Hemoglobin → Methemoglobin

Potassium ferricyanide oxidizes hemoglobin into methemoglobin.

Hb $\xrightarrow{K_3Fe(CN)_6}$ MetHb

5 Methemoglobin → Cyanmethemoglobin

Potassium cyanide reacts with methemoglobin to form cyanmethemoglobin, which has maximum absorption at 540 nm.

MetHb $\xrightarrow{KCN}$ HiCN (CyanmetHb)

6 Measure absorbance at 540 nm

Using a colorimeter or spectrophotometer.



7 Determine hemoglobin concentration

The color intensity (absorbance) is proportional to the total hemoglobin concentration in the sample.

Higher color intensity = Higher Hb concentration

Principle

1. Whole blood is diluted 1:250 with Drabkin's solution (contains potassium ferricyanide and potassium cyanide).
2. RBCs are hemolyzed by Drabkin's solution. Now hemoglobin is oxidized.
3. Hemoglobin is oxidized by potassium ferricyanide into methemoglobin.
4. Then methemoglobin reacts with potassium cyanide to form cyanmethemoglobin, which has maximum absorption at 540 nm.
5. The color intensity measured at 540 nm is proportional to the total hemoglobin concentration in the sample.

Key Reagents and Their Role

Component	Function
Potassium ferricyanide ($K_3Fe(CN)_6$)	Oxidizes Hb to methemoglobin
Potassium cyanide (KCN)	Converts methemoglobin to cyanmethemoglobin
Drabkin's solution	Provides the oxidizing and lysing environment

Important Points

- **Wavelength:** 540 nm
- **Measurement:** Colorimeter / Spectrophotometer
- **Color intensity:** Directly proportional to Hb concentration
- **Dilution ratio:** 1:250

Safety Note

Drabkin's reagent contains highly toxic cyanide. It must be handled with appropriate laboratory precautions and disposed of as hazardous chemical waste.

Continued...

- **1:250 Dilution Ratio in Hemoglobin (Hb) Estimation**

  1:250 means that the blood is diluted to a total of approximately 250 parts.

- **In Drabkin's method:**

  - Blood: 20 µL (0.02 mL)

  - Drabkin's reagent: 5 mL

  - Dilution ratio: (5 /0.02 = 250)

Why is dilution done? To hemolyze RBCs and convert hemoglobin into cyanmethemoglobin so its concentration can be measured accurately at 540 nm.

- **Remember:** 20 µL blood + 5 mL Drabkin's reagent gives approximately a 1:250 dilution

Explanation

- 1. Dilution: Whole blood is mixed with Drabkin's reagent at a dilution of 1:250.
- 2. Hemolysis: The reagent breaks open RBCs and releases hemoglobin into the solution.
- 3. Oxidation: Potassium ferricyanide converts hemoglobin into methemoglobin.
- 4. Cyanide reaction: Potassium cyanide reacts with methemoglobin to form cyanmethemoglobin, a stable colored compound.
- 5. Color measurement: The absorbance of cyanmethemoglobin is measured at 540 nm.
- 6. Calculation: The absorbance is compared with a standard or calibrated reference to determine the hemoglobin concentration.

Important Points

Component	Function
Potassium ferricyanide	Oxidizes Hb to methemoglobin
Potassium cyanide	Converts methemoglobin to cyan-methemoglobin
540 nm	Wavelength used for absorbance measurement
Color intensity	Directly proportional to Hb concentration

- **Safety note:** Drabkin's reagent contains highly toxic cyanide. It must be handled with appropriate laboratory precautions and disposed of as hazardous chemical waste

Drabkin (Cyanmethemoglobin) Method

Procedure – Flowchart

1 Set the blank

- Adjust the spectrophotometer's optical density to zero with blank.



Blank
(Drabkin's solution)

Set OD = 0

2 Transfer standard (1 mL)

- Transfer 1 mL from the ready-to-use standard and read the absorbance.



Read absorbance of standard

3 Prepare sample

- Pipette 5 mL of Drabkin's reagent into a test tube and add 20 µL of whole blood (1:250 dilution).



Drabkin's reagent
5 mL

+

Whole blood
20 µL
(1:250 dilution)

→

Mix thoroughly

4 Mix and allow to stand

- Mix the solution thoroughly and allow it to stand for at least 15 minutes at room temperature to ensure the reaction is complete.



15 minutes
(at room temperature)

5 Measure absorbance

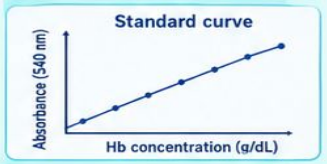
- Use a spectrophotometer to measure the absorbance of the samples and standard, against the blank Drabkin's solution at a wavelength of 540 nm.



Measure at 540 nm

6 Determine concentration

- Calculate the hemoglobin concentration using the absorbance reading, a standard curve, or a formula provided with the kit.



▶

Hb concentration
=
from standard curve
or formula

Key Points

- **Wavelength:** 540 nm
- **Blank:** Drabkin's solution
- **Standard:** Ready-to-use standard
- **Sample:** 5 mL reagent + 20 µL blood (1:250)

Result

Absorbance is proportional to the total hemoglobin concentration in the sample.

Note

Use Drabkin's reagent with care (it contains toxic cyanide). Follow proper laboratory safety and disposal procedures.

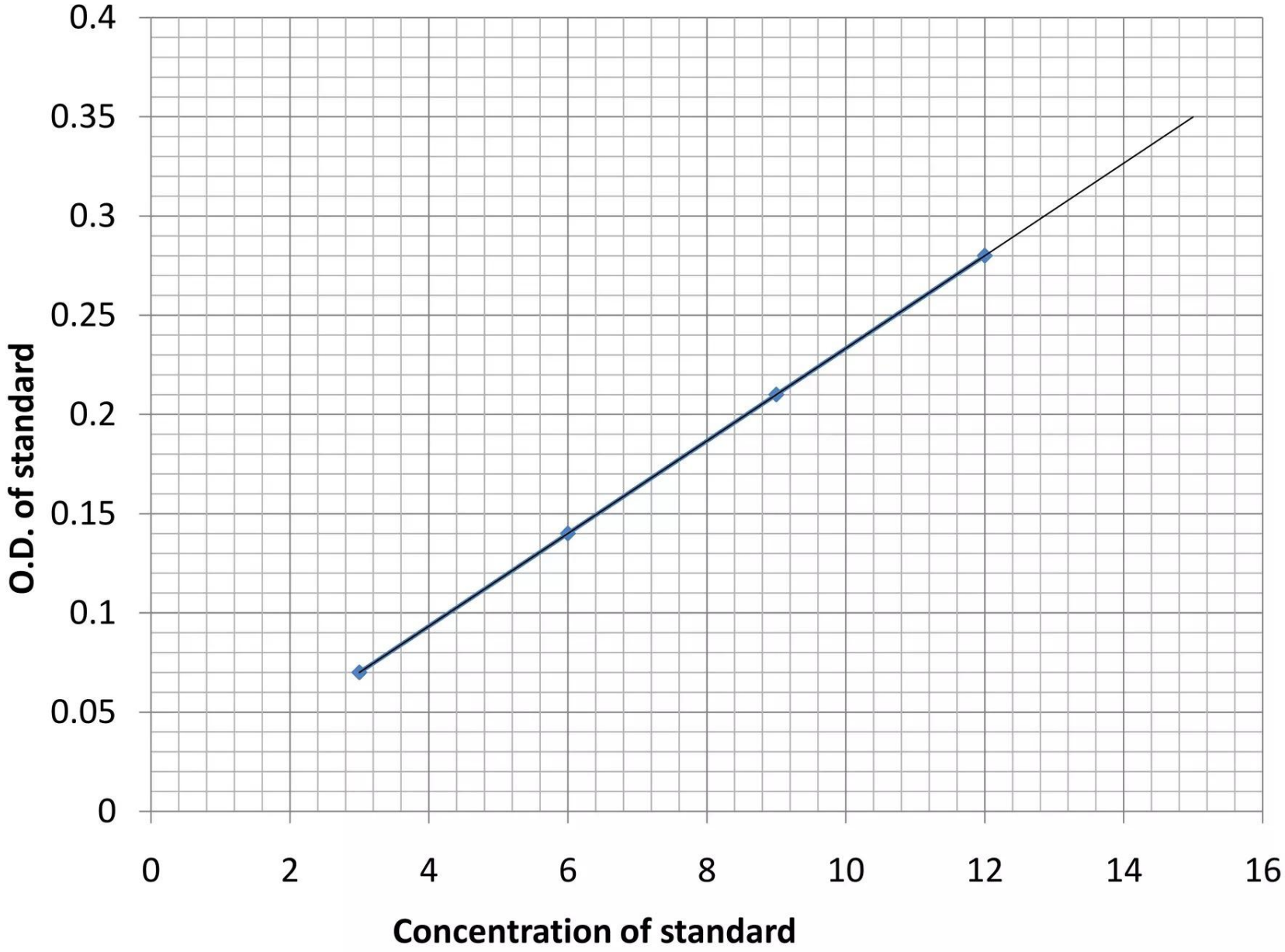
Hemoglobin Calculation Formula:

STANDARD ABSORBANCE RATIO FORMULA

$$\frac{\text{Optical Density of Test (ODtest)}}{\text{Optical Density of Standard (ODstd)}} \times \text{Conc. of Standard (g/dL)} = \text{Hb Concentration (g/dL)}$$

ODtest: Absorbance reading of patient blood sample

ODstd: Absorbance reading of reference standard


0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
O.D. of standard
0
2
4
6
8
10
12
14
16
Concentration of standard

Key Advantages of Drabkin Method

- 1. No Visual Color-Matching Error

The method uses a spectrophotometer to measure absorbance, so there is no need to compare the blood sample's color with a color standard by eye. This reduces subjective errors.

- 2. Stable Color Development

The cyanmethemoglobin solution is stable, and its color does not fade rapidly. Therefore, absorbance readings do not have to be taken immediately after preparation, provided the recommended procedure is followed.

- 3. Reliable Reference Standard

A standardized hemoglobin reference material is available for calibration and comparison, helping laboratories obtain consistent and reliable results.

- 4. Internationally Recognized Reference Method

The cyanmethemoglobin method has traditionally been an internationally recognized reference method for quantitative hemoglobin estimation because of its accuracy and reliability.

- 5. Measures Almost All Forms of Hemoglobin

Most forms of hemoglobin are converted into cyanmethemoglobin, allowing measurement of total hemoglobin. However, sulfhemoglobin is not converted by this method.

Safety Hazard & Disadvantage



Poisonous Chemical Hazard
Potassium Cyanide Toxicity

Drabkin's reagent contains potassium cyanide, a highly toxic chemical substance. Extreme care must be exercised during preparation and measurement.

Safety Precautions:

- Strictly avoid ingestion or aerosol inhalation
- Use automated pipetting; never pipette by mouth
- Dispose cyanide waste according to lab safety rules

Sulphemoglobin Exception

Sulphemoglobin is not converted to cyanmethemoglobin and remains undetected by standard Drabkin's spectrophotometry.

Comparison of Hemoglobin Test Methods

Feature Parameter	Sahli Method	Drabkin Method	Automated Analyzer
Measurement Type	Visual color matching	Spectrophotometer (540nm)	Automated optical detection
Accuracy & Precision	Low (Subjective error)	**High (Reference method)**	Very High
Reagent Chemical	0.1 N Hydrochloric Acid	**Potassium Cyanide**	Proprietary lytic reagents
International Standard	Not standard	WHO Standardized	Calibrated against WHO


THANK YOU